

Polars MariaDB

Leer tabla de base de datos

Referencias

[Documentación API Polars](#)

[Connector-x: Supported Sources & Destinations](#)

```
pip install polars connectorx
```

```
import polars as pl
```

Lectura vía [connectorx](#)

Publicación **VLDB 2022**: [ConnectorX: Accelerating Data Loading From Databases to Dataframes](#)

El código siguiente permite leer los datos desde una tabla en la BD a una tabla Arrow

```
import connectorx as cx
import pyarrow.compute as pc
conn = 'mysql://usuario:contraseña@mariadb:3306/kafka'
query = 'SELECT * FROM songs'
# Leer produciendo una tabla arrow
ta = cx.read_sql(conn, query, return_type='arrow')
print("Tabla Arrow:", ta.schema)
pjta = ta.filter(pc.equal(ta['artist'], 'Pearl Jam'))
print(pjta.select(["album", "title"]))
```

```
Tabla Arrow: ids: int64
title: string
album: string
duration: string
release: int64
artist: string
type: string
filename: string
fechreg: timestamp[us]
pyarrow.Table
album: string
title: string
----
album: [["Ten", "Ten", "Ten", "Ten"]]
title: [["Once", "Even Flow", "Alive", "Jeremy"]]
```

Leer datos de una tabla en un Polars Dataframe indicando el URI de conexión

```
uri = "mysql://usuario:contraseña@mariadb:3306/kafka"
query = "SELECT * FROM songs"
```

```
pl.read_database_uri(query=query, uri=uri)
```

ids	title	album	duration	release	artist	type	filename	fechreg
i64	str	str	str	i64	str	str	str	datetime[μs]
1	"Black Hole Sun"	"Superunknown"	"05:06"	1994	"Soundgarden"	"Rock"	"metal-musics-dataset.csv"	2025-04-09 07:59:48
2	"Smells Like Teen Spirit"	"Nevermind"	"05:01"	1991	"Nirvana"	"Grunge"	"metal-musics-dataset.csv"	2025-04-09 07:59:48
3	"Breed"	"Nevermind"	"03:03"	1991	"Nirvana"	"Grunge"	"metal-musics-dataset.csv"	2025-04-09 07:59:48
4	"Lithium"	"Nevermind"	"04:17"	1991	"Nirvana"	"Grunge"	"metal-musics-dataset.csv"	2025-04-09 07:59:48
5	"Once"	"Ten"	"03:51"	1991	"Pearl Jam"	"Grunge"	"metal-musics-dataset.csv"	2025-04-09 07:59:48
6	"Even Flow"	"Ten"	"04:53"	1991	"Pearl Jam"	"Grunge"	"metal-musics-dataset.csv"	2025-04-09 07:59:48
7	"Alive"	"Ten"	"05:40"	1991	"Pearl Jam"	"Grunge"	"metal-musics-dataset.csv"	2025-04-09 07:59:48
8	"Jeremy"	"Ten"	"05:18"	1991	"Pearl Jam"	"Grunge"	"metal-musics-dataset.csv"	2025-04-09 07:59:48
9	"Forty Six & 2"	"Aenima"	"06:04"	1994	"Tool"	"Metal Progresivo"	"metal-musics-dataset.csv"	2025-04-09 07:59:48
10	"Lateralus"	"Lateralus"	"09:24"	2001	"Tool"	"Metal Progresivo"	"metal-musics-dataset.csv"	2025-04-09 07:59:48

Crear conexión a la base de datos mediante Alchemy

```
import pandas as pd
from sqlalchemy import create_engine
```

```
# Define the connection string
username = 'usuario'
password = 'contraseña'
host = 'mariadb:3306'
database = 'kafka'

# Creating the engine
dbcon = create_engine(f'mariadb+mariadbconnector://{username}:{password}@{host}/{database}', pool_size=10, max_overflow=20)
```

Consultar la base de datos usando la conexión dbcon

SQLAlchemy

`polars.read_database()` requiere una conexión *SQLAlchemy*. Por eso, normalmente es más eficiente, y más simple, usar `polars.read_dabase_uri()`

```
pl.read_database(query="SELECT * FROM songs", connection=dbcon)
```

ids	title	album	duration	release	artist	type	filename	fechreg
i64	str	str	str	i64	str	str	str	datetime[μs]
1	"Black Hole Sun"	"Superunknown"	"05:06"	1994	"Soundgarden"	"Rock"	"metal-musics-dataset.csv"	2025-04-09 07:59:48
2	"Smells Like Teen Spirit"	"Nevermind"	"05:01"	1991	"Nirvana"	"Grunge"	"metal-musics-dataset.csv"	2025-04-09 07:59:48
3	"Breed"	"Nevermind"	"03:03"	1991	"Nirvana"	"Grunge"	"metal-musics-dataset.csv"	2025-04-09 07:59:48
4	"Lithium"	"Nevermind"	"04:17"	1991	"Nirvana"	"Grunge"	"metal-musics-dataset.csv"	2025-04-09 07:59:48
5	"Once"	"Ten"	"03:51"	1991	"Pearl Jam"	"Grunge"	"metal-musics-dataset.csv"	2025-04-09 07:59:48
6	"Even Flow"	"Ten"	"04:53"	1991	"Pearl Jam"	"Grunge"	"metal-musics-dataset.csv"	2025-04-09 07:59:48
7	"Alive"	"Ten"	"05:40"	1991	"Pearl Jam"	"Grunge"	"metal-musics-dataset.csv"	2025-04-09 07:59:48

ids	title	album	duration	release	artist	type	filename	fechreg
i64	str	str	str	i64	str	str	str	datetime[μs]
8	"Jeremy"	"Ten"	"05:18"	1991	"Pearl Jam"	"Grunge"	"metal-musics-dataset.csv"	2025-04-09 07:59:48
9	"Forty Six & 2"	"Aenima"	"06:04"	1994	"Tool"	"Metal Progresivo"	"metal-musics-dataset.csv"	2025-04-09 07:59:48
10	"Lateralus"	"Lateralus"	"09:24"	2001	"Tool"	"Metal Progresivo"	"metal-musics-dataset.csv"	2025-04-09 07:59:48

Escribir en tabla de la base de datos

```
# Leer datos de archivo CSV local
dfpl = pl.read_csv("Eurostar/stations.csv")

# Volcar en la base de datos
dfpl.write_database(table_name='stations', connection=dbcon, engine='sqlalchemy',
if_table_exists = "replace")
```